

CSC310 Data Structures and Algorithm Analysis (Dr. Ng)

Programming Assignment: Basic Tic-Tac-Toe

Write a program to play the 3 by 3 “Tic-Tac-Toe” board game with the following characteristics:

1. The program would allow two human players and capture their names. Each name consists of first name and last name with a space between them. A valid player's name consists of only alphabet letters. Players' names are entered, one player at a time per line. Each player's name is input on the same line, but separately one player at a time per line.
2. The full player's name will be used in displaying game statistics (#8 below). However, only the first names of players are to be used in subsequent prompt messages (#6 below).
3. The program will allow first player to choose which piece ('x' or 'o', **both lower case**) to play with, and the second player will play with the other piece. Once the game starts, the players keep their pieces throughout the game.
4. **The player with the shorter first name will start the first game. In case of a tie, the player that entered the name first in #1 will start the game.**
5. The game board should look **EXACTLY** like the one below and is referenced by numbers 1 to 9:

```
+---+---+---+
| 1 | 6 | 2 |
+---+---+---+
| 7 | 5 | 8 |
+---+---+---+
| 3 | 9 | 4 |
+---+---+---+
```

6. The program will prompt user's move accordingly using **ONLY** the first name.
7. The program would redraw the board **ONCE** after each valid move.
8. A winning configuration is when the 3 pieces horizontally, vertically or along the two diagonals are same. If one player wins, **ALL** possible winning configurations will be displayed on the board in upper case “O” or “X”.
9. At the end of each game, the program would display the total number of games played and the number of games each player (with the full name: first name and last name) wins, loses and draws.
10. The loser of the previous game will start the next game. In case of a draw game, the players will alternate who starts the next game.
11. The program would allow multiple games to be played until the players select to quit at the end of any game.

You should use C++ for this assignment. You cannot call any library functions except those dealing with I/O and string handling functions. Sophisticated graphic display is NOT necessary, and it wouldn't get you extra points. All that is required is **a neat,**

consistent and reasonable user-friendly user interface. Pay attention to space out your output and prompts.

Your program should be executed interactively on your computer. Please test your program thoroughly and make sure it is free of errors. Points will be deducted for each error that I discovered for you.

In addition, your program should account for all possible user input errors (Account for ALL possible errors) and prompt the user with appropriate error messages.

Please also pay attention to **program design, human-computer interface design, correct grammar and punctuation, and follow all the good programming practice.** Points will be deducted points based on these criteria.

Before you start coding, you should think carefully about the data structures (with justifications) used to

- keep track of player information and statistics**
- keep track of the current game status**

**Submit: Program and Data Structure design discussion
Hierarchical diagrams/ Structure charts/Program Decomposition Diagram
Procedure specification
Program listing
Testing cases with justifications and expected output
Actual results of test cases, screen shots
Limitation of current version and future works**

Optional Extensions Parts A and B:

PART A:

Implement a computer player (from dummy or intelligent, different levels) to play the game.

PART B:

Modify your program to play the standard 7 by 7 two players "CONNECT-4" board game.